

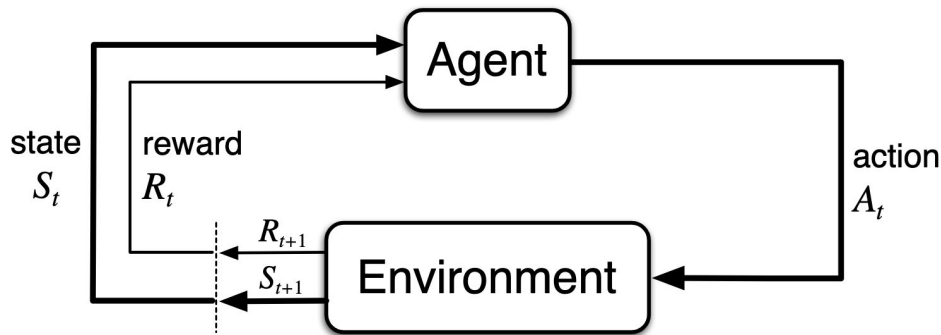
Characterizing Reinforcement Learning Environments by their Exploration Requirement

Projet MIND 2025-2026

Antipin Vladislav & Oundjian Milos

Introduction - Reinforcement Learning

- ❑ **Agent:** interacts with environment
- ❑ **Environment:** gives result of action
 - ❑ (state, action) $\rightarrow$ (next state, reward)
- ❑ **Policy:** decision making strategy
 - ❑ state $\rightarrow$ action



Environments defined by MDPs

$$(\mathcal{S}, \mathcal{A}, T, R, \gamma)$$

Goal to find the best policy $\pi^* = \arg \max_{\pi} \mathbb{E} \left[\sum_{t=0}^T \gamma^t r(s_t, a_t) \right]$

Transitions: (state, action, reward, next state)

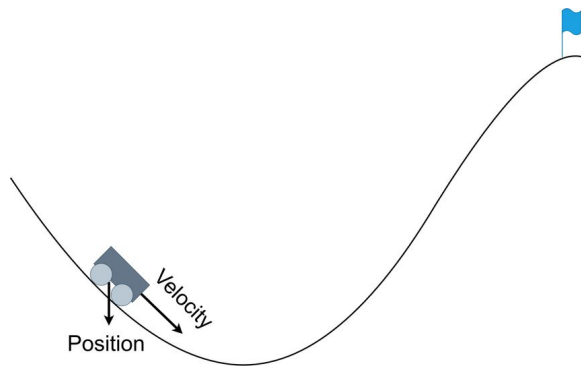
Exploration-Exploitation Tradeoff

Exploration

- ❑ Discover new states
- ❑ Improve coverage of environment
- ❑ Too much = Waste time

Exploitation

- ❑ The agent follows what it knows works
- ❑ Too much = Non-optimal policy

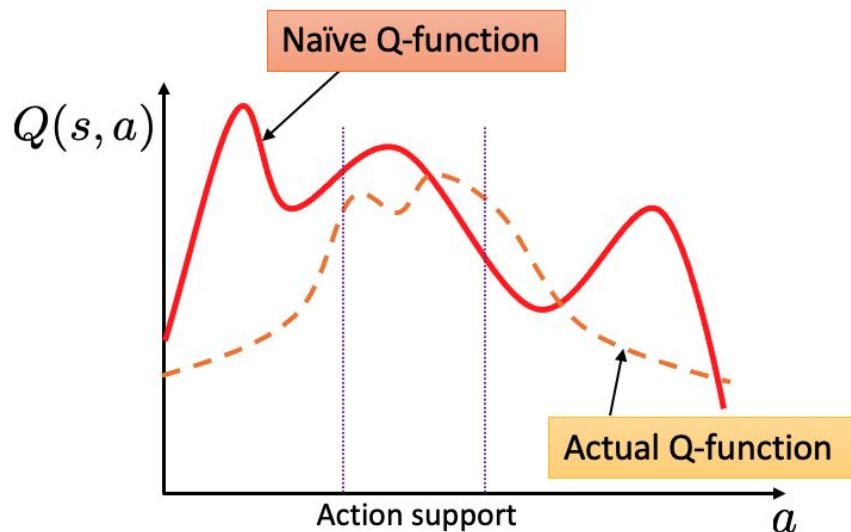


Offline RL Challenges

- ❑ Fixed replay buffer
- ❑ Unable to access environment

Issues:

- ❑ The learned policy may deviate from the dataset distribution
 - ❑ Distributional Shift
 - ❑ Out of distribution
- ❑ Limited Exploration



How can RL environments be characterized according to the level of exploration required for effective offline learning?

Environments

Continuous 1D action space

❑ CartPole

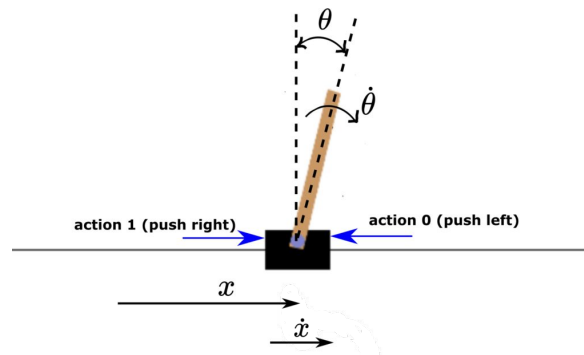
- ❑ 4D: (position, velocity, angle, angular velocity)
- ❑ force

❑ Pendulum

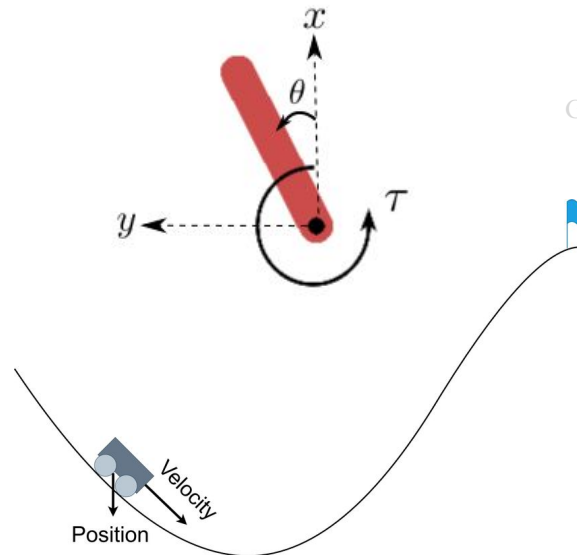
- ❑ 2D: (angle, angular velocity)
- ❑ Torque

❑ MountainCar

- ❑ 2D: (position, velocity)
- ❑ force



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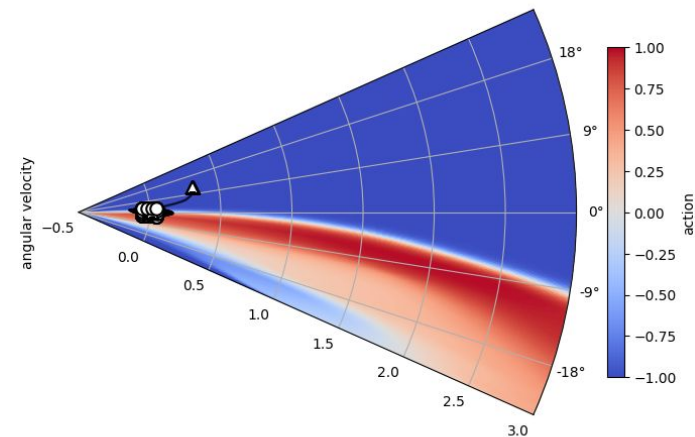


Gymnasium documentation

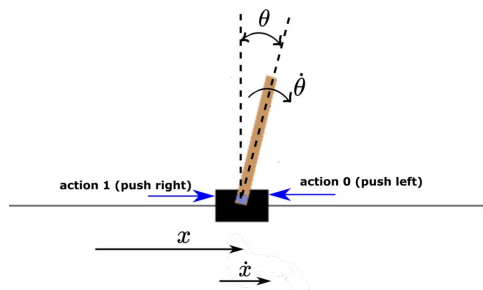
Zolfagharian et al., 2023

Online-Trained Exploitation Policies

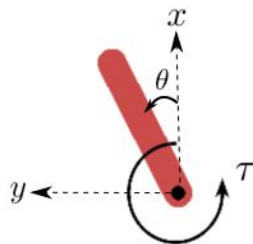
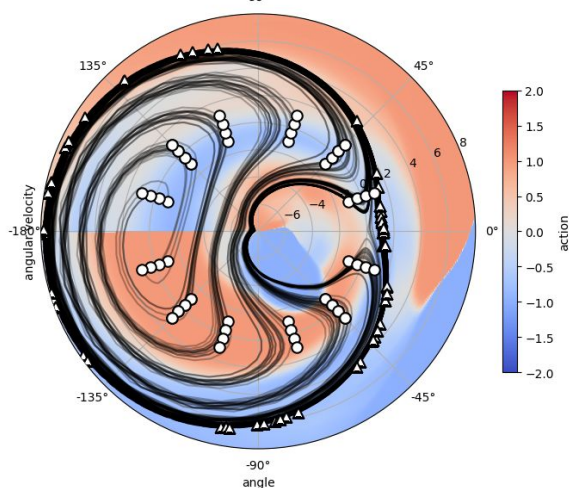
CartPoleContinuous-v1



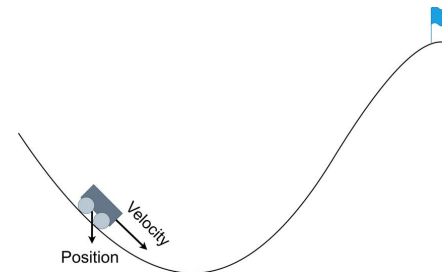
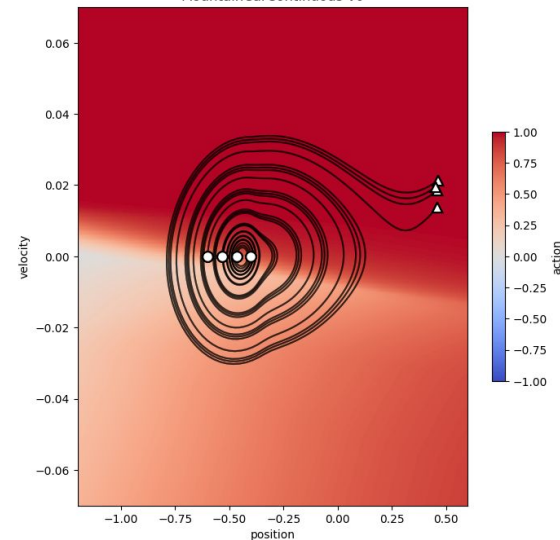
angle



Pendulum-v1



MountainCarContinuous-v0

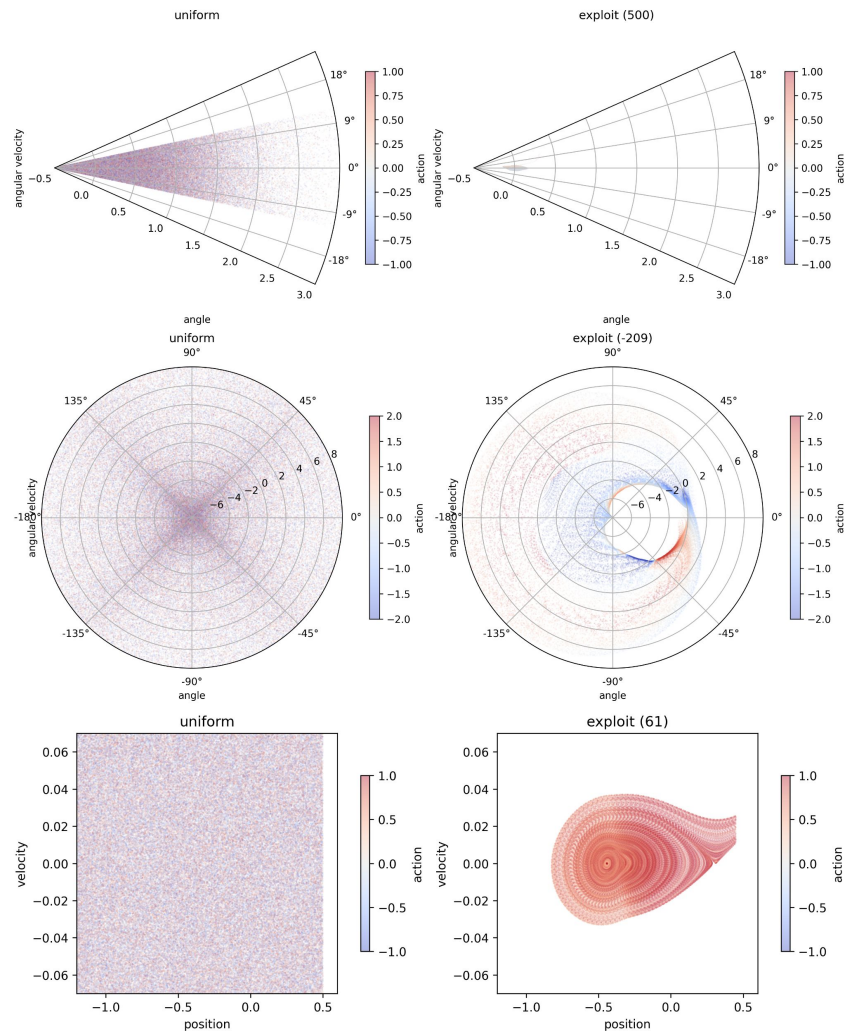


Replay Buffer Collection

- ❑ **Exploration: Uniform Sampling**
 - ❑ Uniformly sample state action pairs in the environment
- ❑ **Exploitation: Policy Transitions**
 - ❑ Run episodes on the environment and collect transitions.

Then mix in proportions:

- ❑ 0-100% uniform exploration



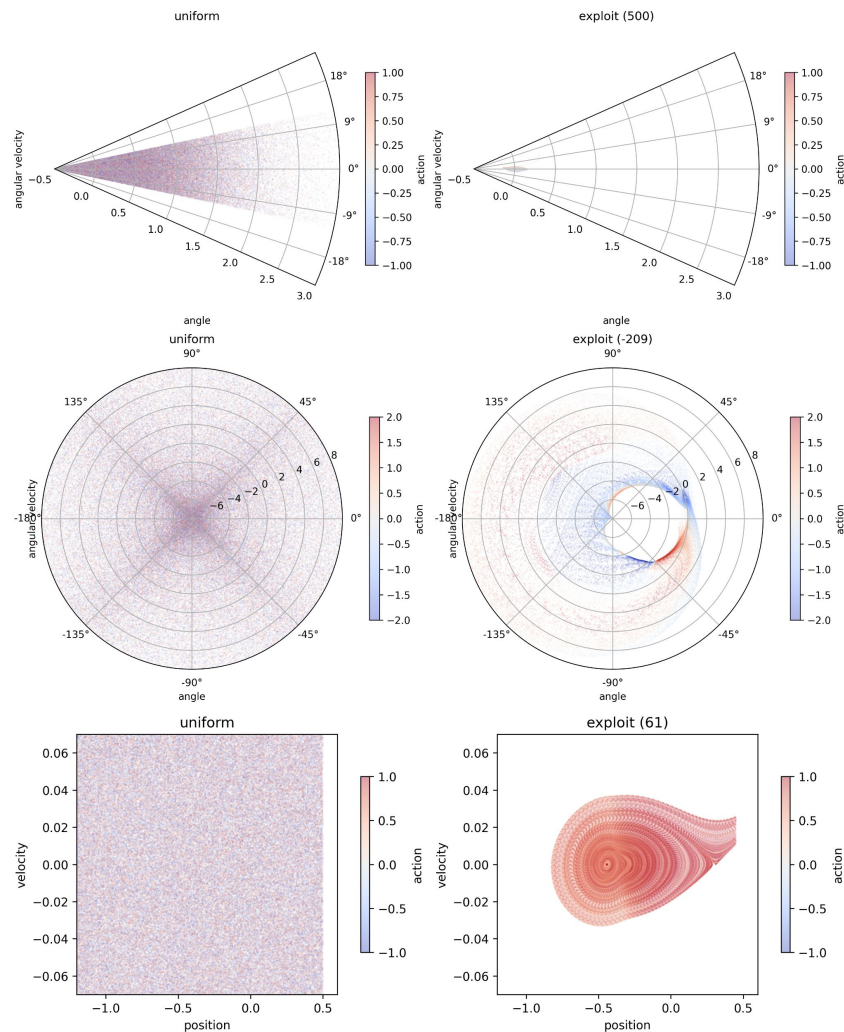
Collected Replay Buffers

❑ Exploration:

- ❑ covers **relevant and irrelevant** regions

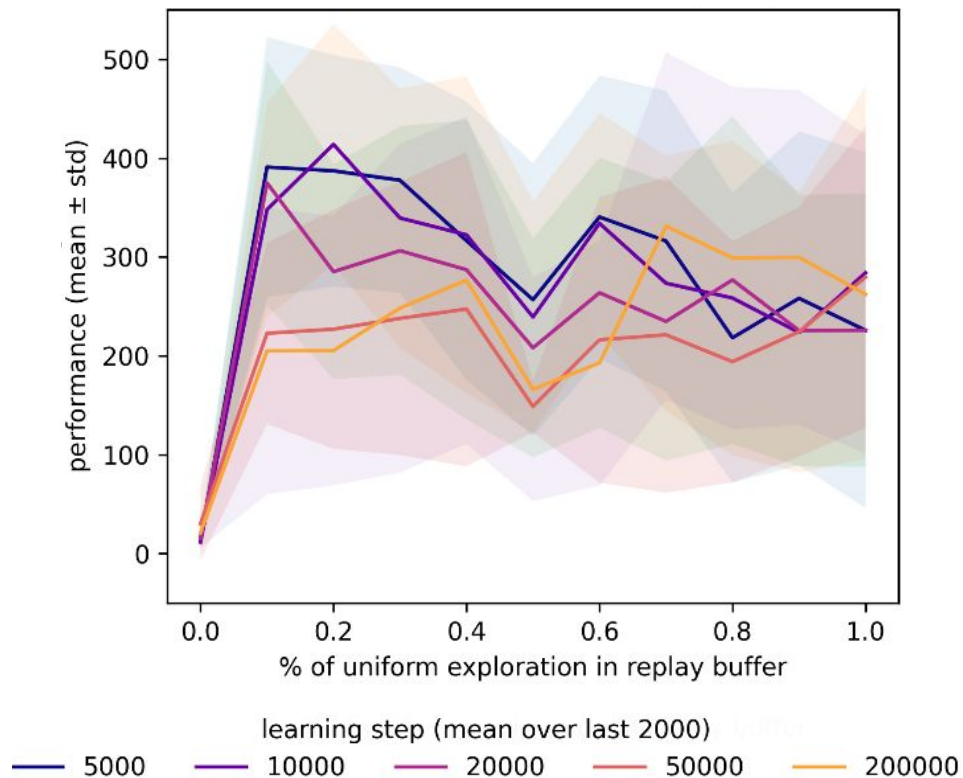
❑ Exploitation:

- ❑ covers **narrow region** in state-action space
- ❑ CartPole:
 - ❑ small angle and angular velocity
 - ❑ actions are close to zero
- ❑ Pendulum:
 - ❑ highest density around equilibrium
- ❑ MountainCar:
 - ❑ negative actions never tried



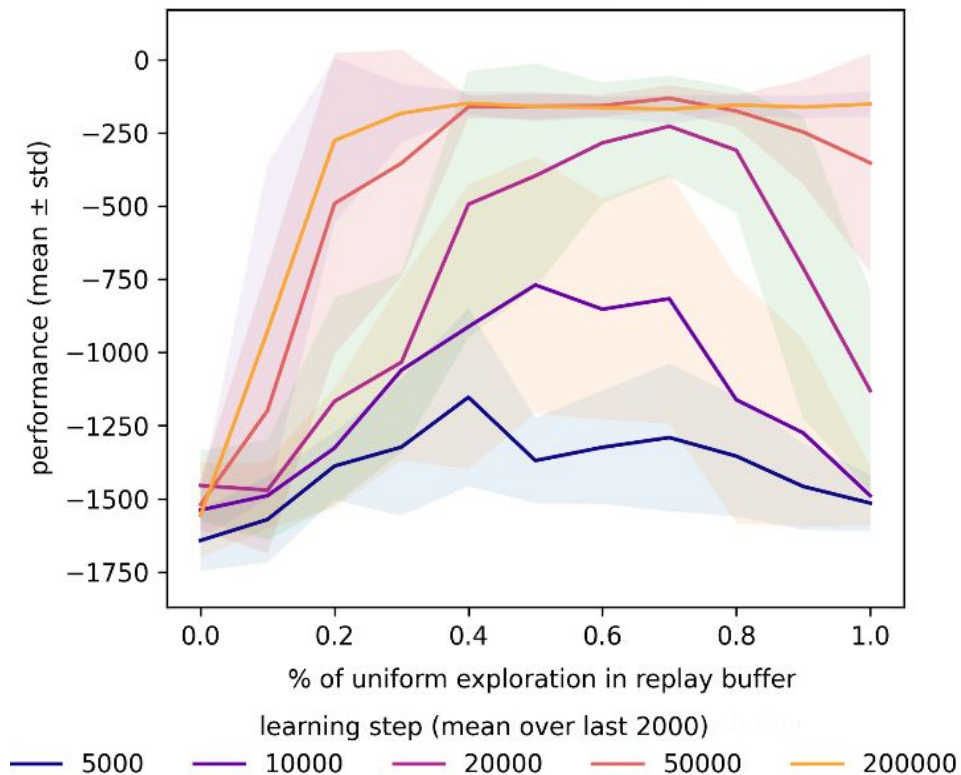
CartPole

- ❑ $\geq 10\%$ exploration $\rightarrow$ plateau
- ❑ 5K steps
- ❑ Overtraining for low exploration
- ❑ Interpretation:
 - ❑ exploration covers situations near falling
 - ❑ exploitation doesn't



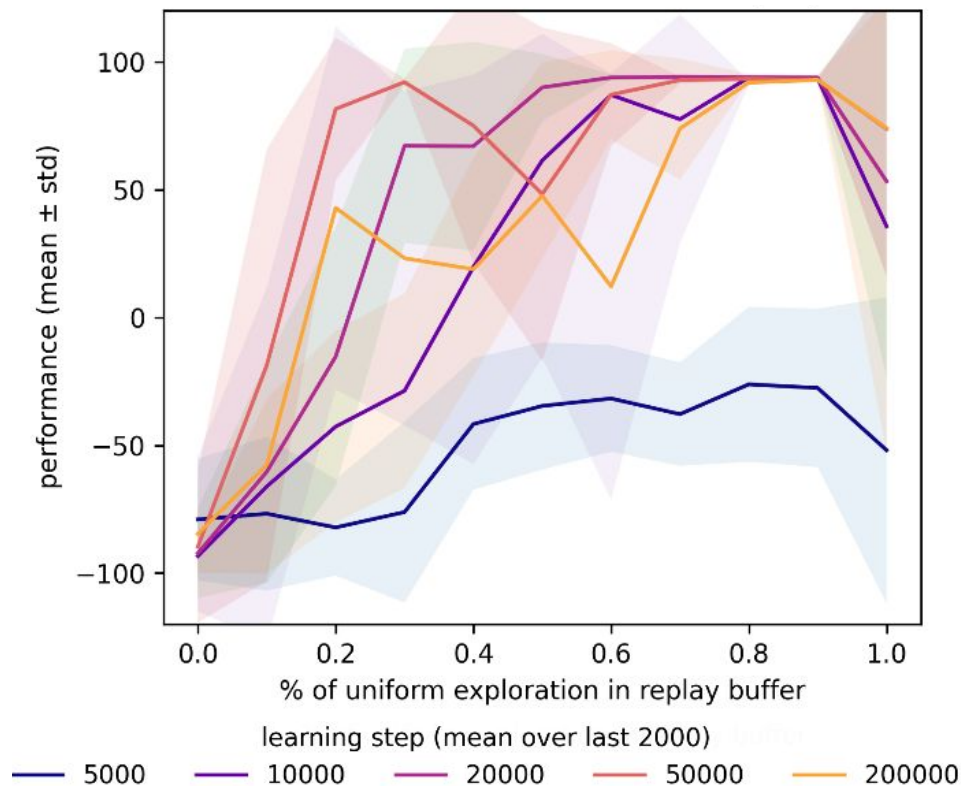
Pendulum

- ❑ $\approx 50\%$ exploration $\rightarrow$ bell curve
- ❑ 20K steps
- ❑ Longer training compensates lack of exploitation $\rightarrow$ plateau
- ❑ Interpretation:
 - ❑ exploration is necessary for recovery
 - ❑ exploitation helps learn balancing faster



MountainCar

- ❑ $\geq 20\%$ exploration $\rightarrow$ plateau
- ❑ 10-20K steps
- ❑ Not trapped at zero performance
- ❑ Interpretation:
 - ❑ exploration helps find a robust way to gain momentum
 - ❑ exploitation and exploration cover the goal region



Conclusion

- ❑ **Uniform exploration** is consistently **beneficial** across all environments
- ❑ **CartPole** needs only a **10% of exploration and learns quickly**, sensitive to overtraining
- ❑ **Pendulum** requires a **balanced mix** to learn the precise control
- ❑ **MountainCar** needs **at least 20% of exploration** to learn the moment building
- ❑ **Uniform sampling: unrealistic and wastes transitions** for irrelevant regions of state-action space
- ❑ Proposed amelioration: **explore around the optimal policy instead** (see the report)

References

1. Reinforcement Learning: An Introduction — Richard S. Sutton and Andrew G. Barto, 2nd edition, MIT Press, 2018.
2. Gymnasium documentation
3. Zolfagharian, A., Abdellatif, M., Briand, L., Bagherzadeh, M., & Ramesh, S. (2023). A Search-Based Testing Approach for Deep Reinforcement Learning Agents. IEEE Transactions on Software Engineering.
4. BAIR Blog – Offline Reinforcement Learning: How Conservative Algorithms Can Enable New Applications
5. Aleksandar Haber Blog

Testing MountainCarContinuous-v0:
[(34.93184280395508, <bbrl.utils.replay_buffer.ReplayBuffer object at 0x134044430>), (60.57007598876953, <bbrl.utils.replay_buffer.ReplayBuffer ob
skipped best policy
intermediate policy (60.57007598876953)...

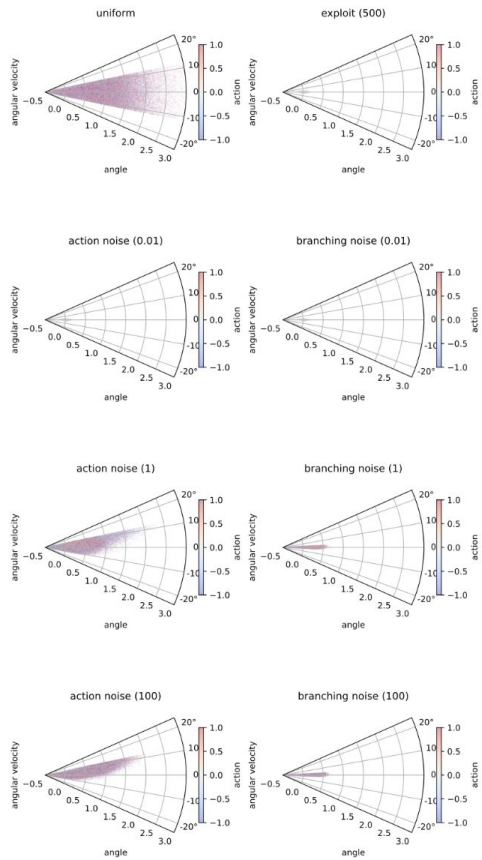
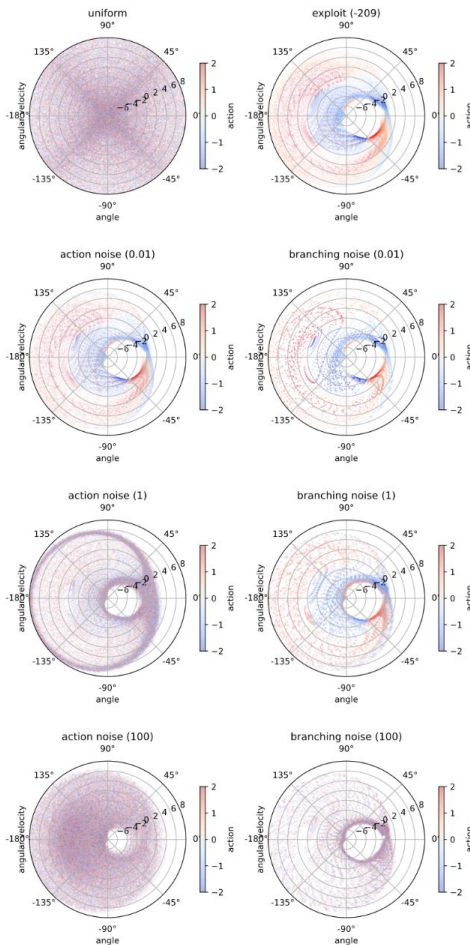
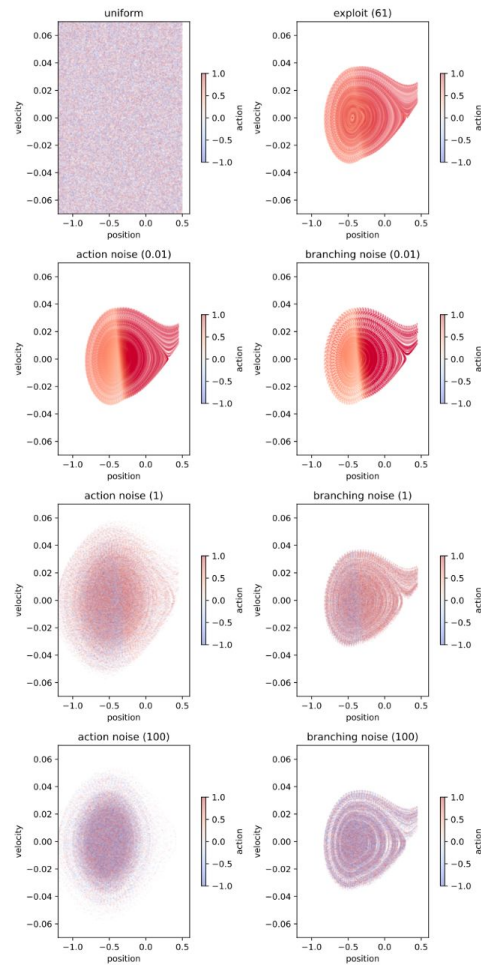
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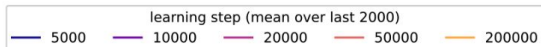
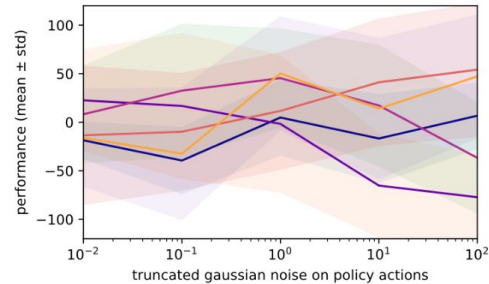
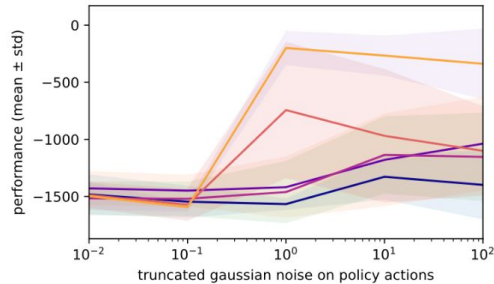
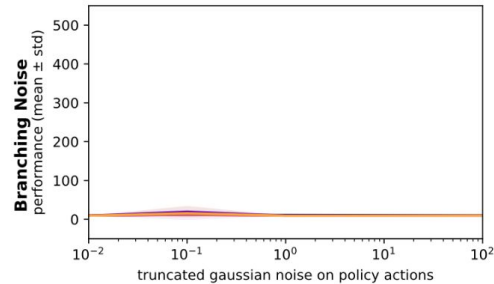
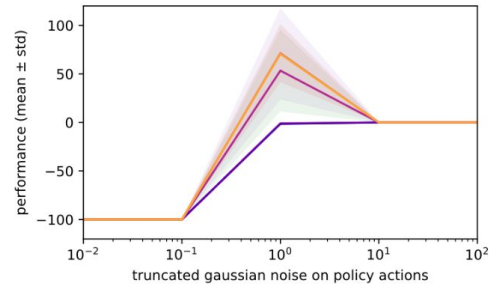
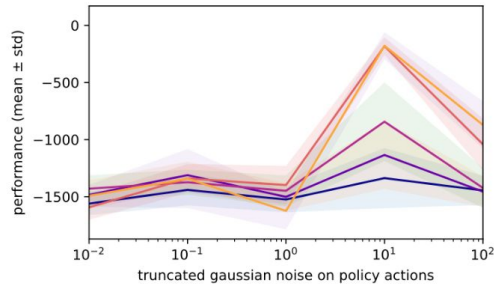
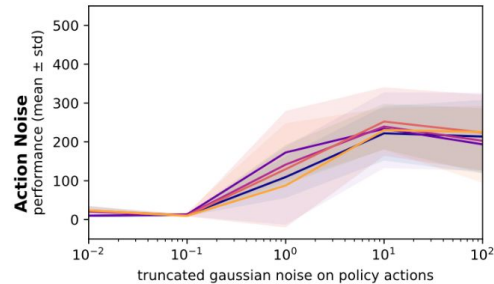
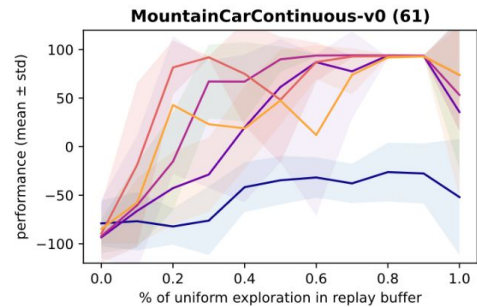
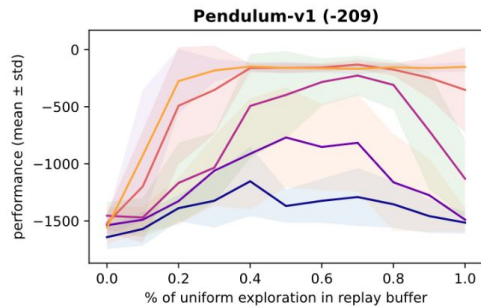
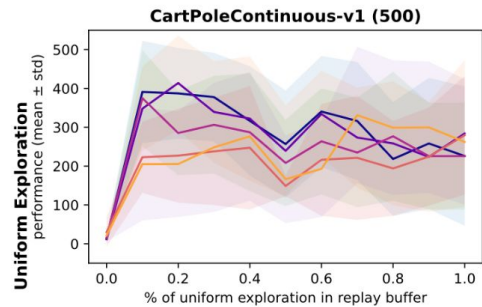
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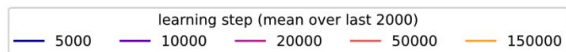
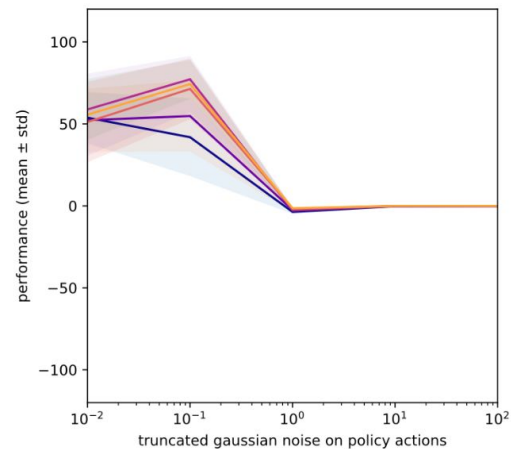
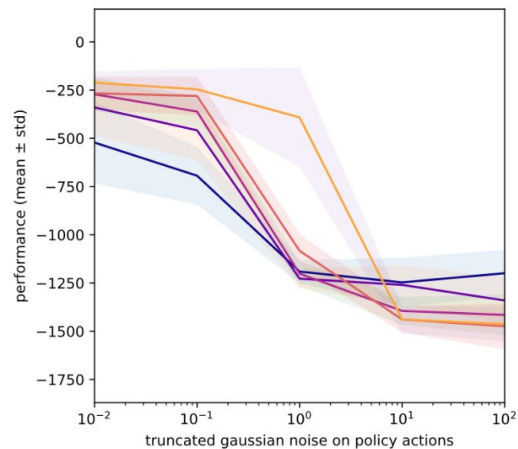
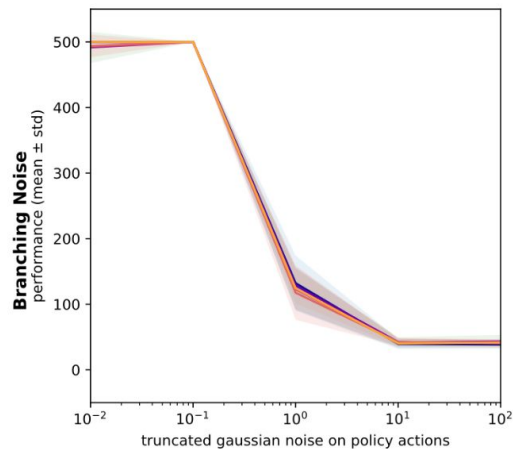
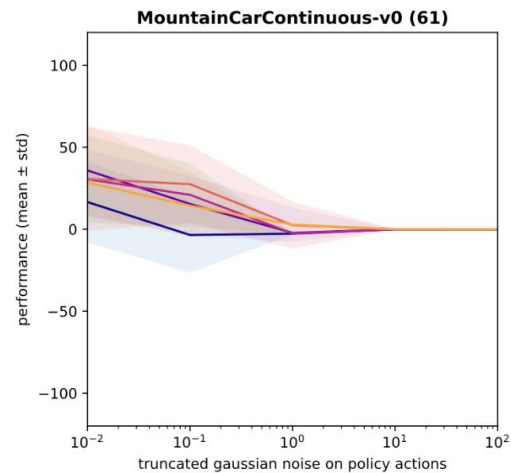
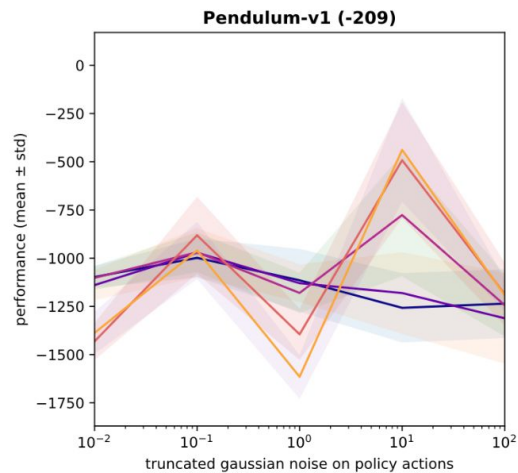
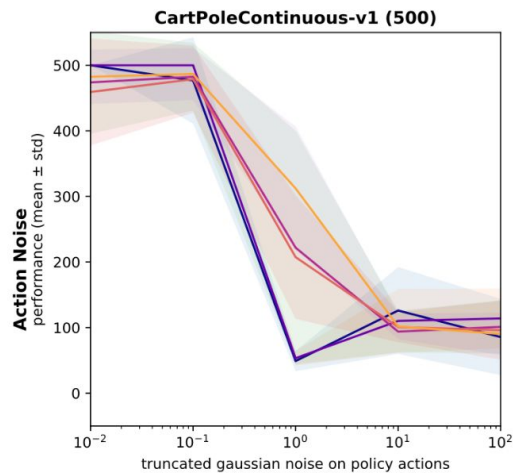
APPENDIX

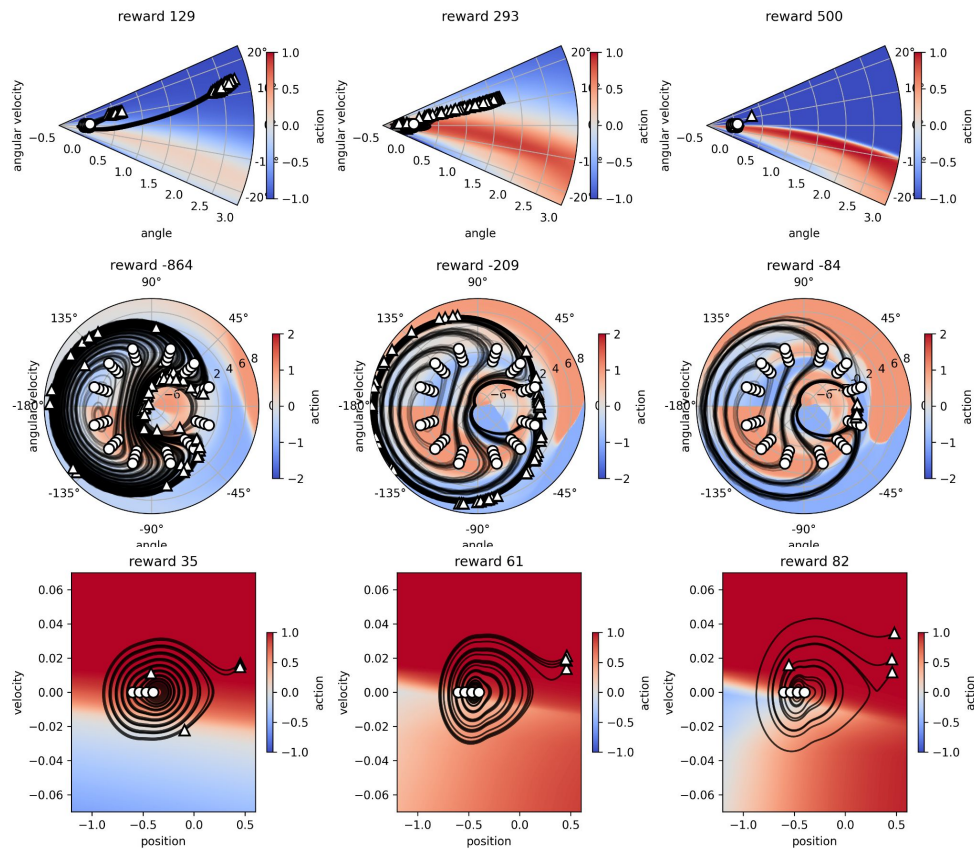
	CartPoleContinuous-v1	Pendulum-v1	MountainCarContinuous-v0
State Space	Cart Position: $[-4.8, 4.8]$ Cart Velocity: $]-\infty, \infty[$ Pole Angle: $[-0.418, 0.418]$ Ang Velocity: $]-\infty, \infty[$	$\cos(\theta)$: $[-1, 1]$ $\sin(\theta)$: $[-1, 1]$ Ang Velocity: $[-8, 8]$	Car Position: $[-1.2, 0.6]$ Car Velocity: $[-0.07, 0.07]$
Action Space	Force: $[-1, 1]$	Torque: $[-2, 2]$	Force: $[-1, 1]$

Table 5.1: State and Action spaces of the different environments used in our experiments.

(a) CartPoleContinuous-v1**(b) Pendulum-v1****(c) MountainCarContinuous-v0**

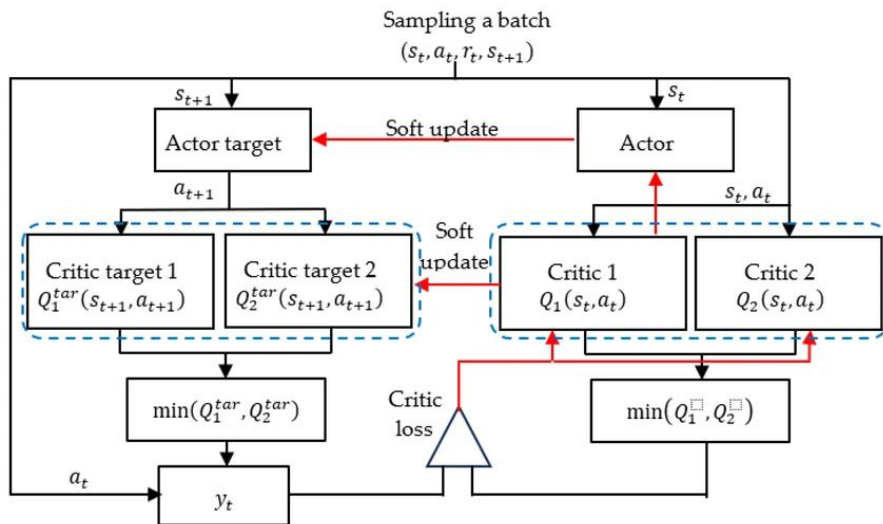






TD3 RL Algorithm

- ❑ Actor Critic Algorithm
 - ❑ Issue: Overestimation bias (is it specific to it?? no)
- ❑ Idea 1: Two critics and take the minimum
- ❑ Idea 2: Delayed actor updates for better stability
- ❑ Idea 3: Target policy smoothing



Markov Decision Process (MDPs)

- ❑ How we characterize environments
- ❑ State Space, Action Space, Transition Function, Reward Function, Discount Factor
 $(\mathcal{S}, \mathcal{A}, T, R, \gamma)$
- ❑ Transitions: (state, action, reward, next state) building block of replay buffer

